



Q1. I am an engineering student and want to do a project on Linux platform. Please suggest.

Palu Patil
Through e-mail

A1. An interesting way to learn and build projects on Linux platform is to use Raspberry Pi. It is a wonderful ARM microcontroller-based development board that can run Raspbian, Risc OS, Arch, Raspbmc and Pidora Linux on it. With the availability of various GPIO pins, you can control any hardware and develop innovative applications. The Raspberry Pi Foundation recommends Python as a language for learners. However, any language which will compile for ARMv6 can be used with Raspberry Pi. So you can use Python, C, C++, Java, Scratch and Ruby, all of which come installed by default on Raspberry Pi. EFY is regularly publishing articles on Raspberry Pi since April 2013.

Q2. Thanks for answering our query on MATLAB colour detection in July issue. We have been able to detect colours. Can human face also be detected with the same algorithm?

Pooja
Through e-mail

A2. No, you will have to use different algorithm for that. Face detection has been regarded as one of the most complex and challenging problems in the field of computer vision due to large variations caused by differences in facial features, lighting and expression.

There are different types of algorithms that can be used for face detection. With MATLAB you can use Viola-Jones algorithm. Cascade-

ObjectDetector function in MATLAB uses Viola-Jones algorithm to detect human faces, noses, eyes, mouth or upper body. You can also use the trainCascadeObjectDetector function to train a custom classifier to use with this System object.

Q3. The solder I am using mentions 60/40 on its cover. What does it mean? What are the different types of solders available in the market?

Bhaskar Pandey
Through e-mail

A3. Solder is a metallic alloy that, when melted with a heated soldering iron, can join metallic sur-

same applies to solidification too. The third composition (62/36/2) is used for professional-grade products. For regular electronics works 63/37 is recommended.

These compositions are not used in European countries where lead-free soldering is a primary requirement. The European Union now requires all commercially-available electronics to use lead-free components, including solder. Lead is used to reduce the melting point of the alloy. With lead-free coming in, different metals need to be used.

Lead-free solders are a bit harder to use as these do not flow as nicely or at as low a temperature as lead-

TABLE I
Different Compositions

Type	Tin (Sn)	Lead (Pb)	Silver (Ag)	Melting temperature
60/40	60%	40%	0%	183-190°C
63/37	63%	37%	0%	183°C
62/36/2	62%	36%	2%	179°C

TABLE II
Lead-Free Solder Compositions

Sn96/Ag4
Sn96.5/Ag3.5
Sn93.6/Ag4.7/Cu1.7
Sn95.2/Ag4/Cu0.8
Sn95.2/Ag3.9/Cu0.9
Sn95.2/Ag3.8/Cu1
Sn95.5/Ag3.5/Cu1
Sn96.2/Ag3/Cu0.7
Sn96.5/Ag3/Cu0.5
Sn96.2/Ag2.5/Cu0.8/Sb0.5



Fig. 1: 60/40 solder wire

faces together. The most common solder compositions used for electronic circuits have tin (Sn), lead (Pb) and silver (Ag) elements. Table I shows different compositions and their melting temperatures.

The most widely used composition in India is 60/40 (refer Fig. 1). This composition doesn't melt on a specific temperature. It melts in a range of temperatures as shown in Table I. 63/37 is more modern, and it melts at a specific temperature. The

based solders. When lead-based solder is used, good joints can immediately be seen as these are shiny and convex shaped. And

with very little practice, you can master good soldering. However, in case of lead-free solder, good joints are not shiny, and it is very difficult to visually detect a bad joint. Table II shows some lead-free solder compositions.

Answers compiled by EFY technical editor, Ankit Gupta. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: editse@efy.in) and should include name and address of the sender